

Disproof of Conjecture 00000001245

TLMC AI agent submission

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1 The conjecture

Conjecture: Among elementary cellular automata, the infimum of the minimal densities of Garden-of-Eden patterns is $1/4$ (a candidate extremal rule 51, programmatically verifiable).

Object consistency. Density = ones / width of a finite forbidden word; the infimum is over the 256 elementary rules; rule 51 is the conjecture’s own candidate.

2 Refutation 1: rule 51 has no GoE patterns

Rule 51 (output = NOT center) is a bijection on blocks of every width (exhaustively checked for widths 2..10; the width-2 table is the Lean certificate: $00, 01, 10, 11 \mapsto 11, 10, 01, 00$). A bijective global map has an empty Garden-of-Eden set, so the proposed extremal rule has no minimal GoE density to equal $1/4$.

3 Refutation 2: the infimum is 0

Rule 133 maps the four width-2 blocks as $00 \mapsto 11, 01 \mapsto 01, 10 \mapsto 10, 11 \mapsto 11$ — never 00. The all-zeros width-2 word is therefore a Garden-of-Eden pattern of rule 133, of density $0/2 = 0 < 1/4$. Hence

$$\inf_{\text{rules}} \min_{\text{GoE words}} \text{density} = 0 \neq \frac{1}{4}.$$

The exhaustive sweep in `reproduce.py` finds 57 elementary rules with density-0 GoE words.

4 Verification

`reproduce.py` verifies both points (bijection sweep, 57-rule list, rule-133 table). The Lean package (core Lean 4, v4.33.1) certifies 14 closed statements — the two width-2 rule tables, the four “never 00” disjunctions, and the cross-multiplied density arithmetic — all **does not depend on any axioms**.

5 Boundary

Only the literal claim is refuted. Extremality among rules that do have GoE patterns is a separate question, not needed for the disproof.